

JTAG

JOINT TEST ACTION GROUP (JTAG) [IEEE 1149.1 std]

NEED OF JTAG :

- To minimize the no. of ports from top level.
- To achieve boundary scan.

To minimize no. of ports from top level

LIST OF SOME IMPORTANT SIGNALS :

TM

edt_bypass_en

int_mode

ext_mode

int_ltest_en

ext_ltest_en

~~mbist_bypass_en~~

mem_bypass_en

These are just a list of important signals.
There will be many more signals like these.

If all these signals are to be controlled from top-level, then the no. of top level ports will increase drastically.

So we use JTAG to internally control these signals.

By using JTAG we need only 5 ports (TMS, TCLK, TRST, TDI, TDO) to control all the above signals.

Using JTAG, we can control only static signals (signals that are constant throughout the run)

Example :

- **edt_byass_en** => static signal

It will either be 0 or 1 for the complete run. It can be controlled using JTAG

- **edt_update** => non static / dynamic signal

For the same run, during load it will be 1, for the rest of the procedures it will be 0. This can't be controlled using JTAG. We have to control from top level.

- **TM** => static signal

TM = 1 for the complete run

It can be controlled using JTAG

- **SE** => non static/ dynamic signal

SE = 1 => during load and shift

SE = 0 => during capture.

It can't be controlled using JTAG. We have to control it from top level.

JTAG ARCHITECTURE

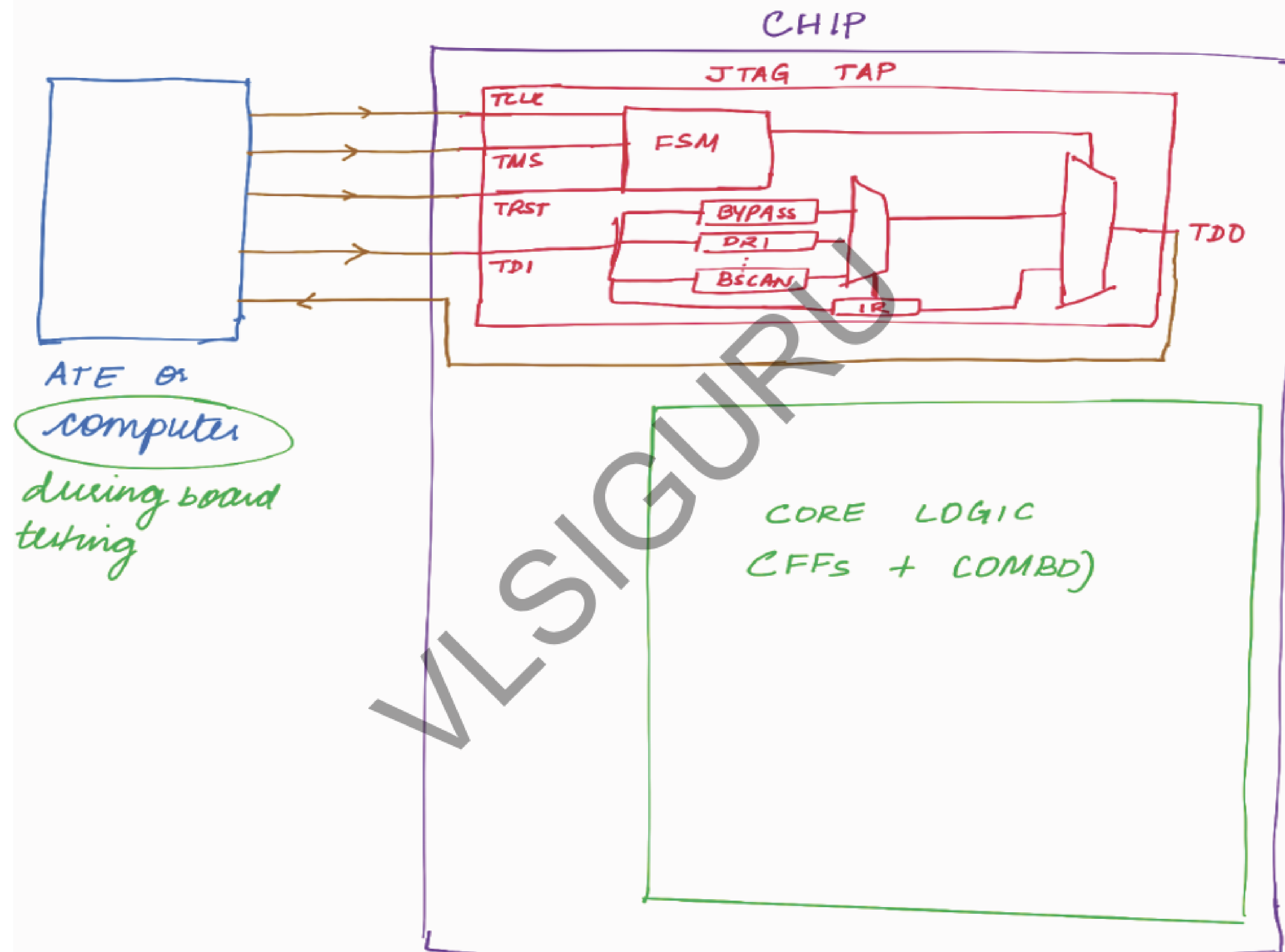
JTAG consists of

- A TAP [Test Access Port]
- A TAP controller [16 state FSM]
- An instruction register [IR]
- A group of test data registers [TDRs]

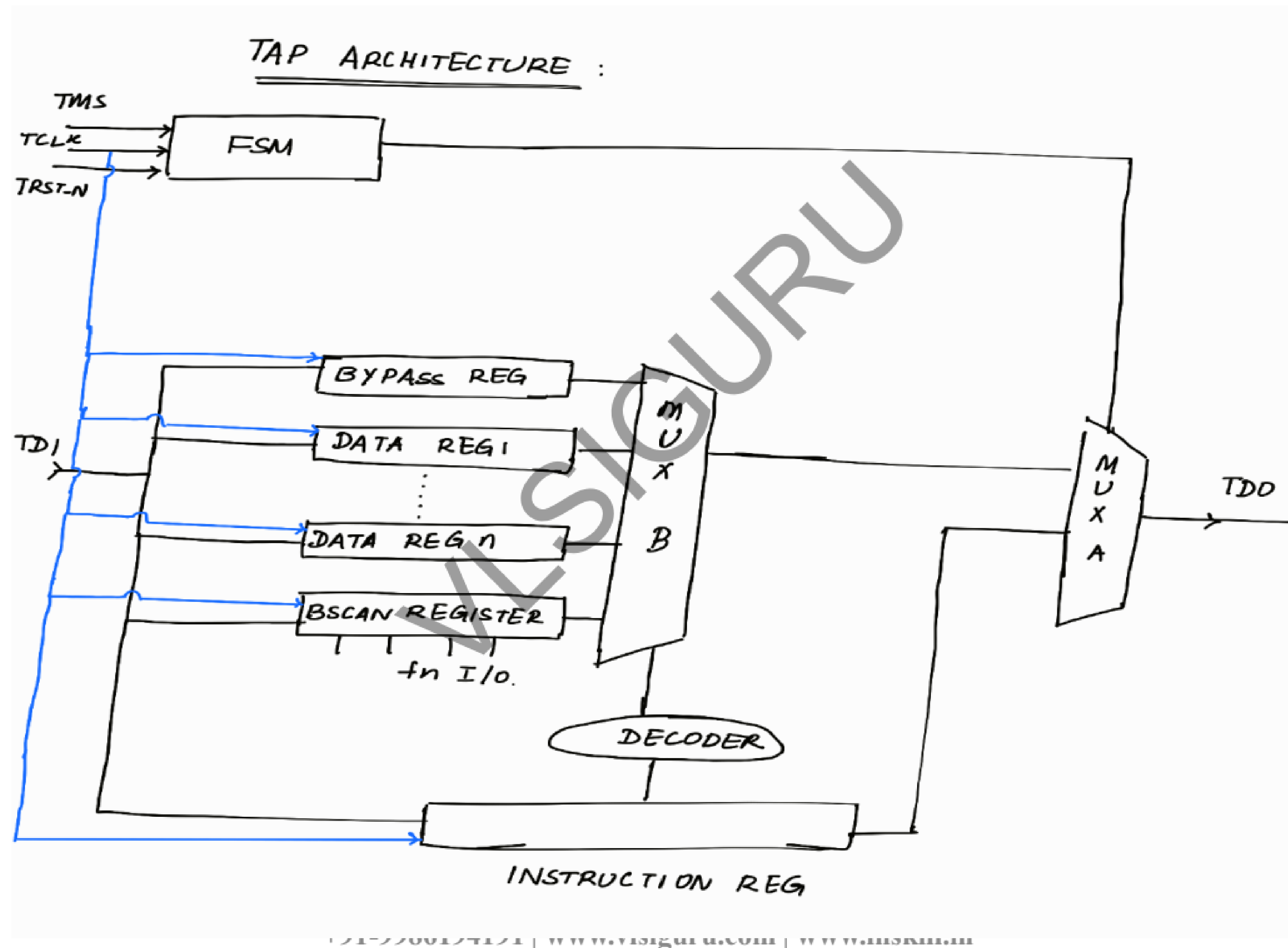
Test Access Port (TAP)

It is the interface used for JTAG control. The IEEE standard defines four mandatory TAP signals and one optional TRST signal.

1. **TDI (Test Data Input)** – Serial input for test instruction and data
2. **TDO (Test Data Output)** – Serial output for test instruction and data
3. **TCK (Test Clock)** – It is the clock to the test data registers (TDR), instruction register (IR).
4. **TMS (Test Mode Select)** – It controls the TAP controller's state transitions.
5. **[Optional] TRST (Test Reset)** – Provides for asynchronous initialization of the TAP controller.



TAP ARCHITECTURE



TAP ARCHITECTURE (Contd...)

Mux A : to select between IR and DR

select line : from JTAG FSM (TAP controller)

When FSM is in any one of the IR states, it will select IR else it will select DR

Mux B :

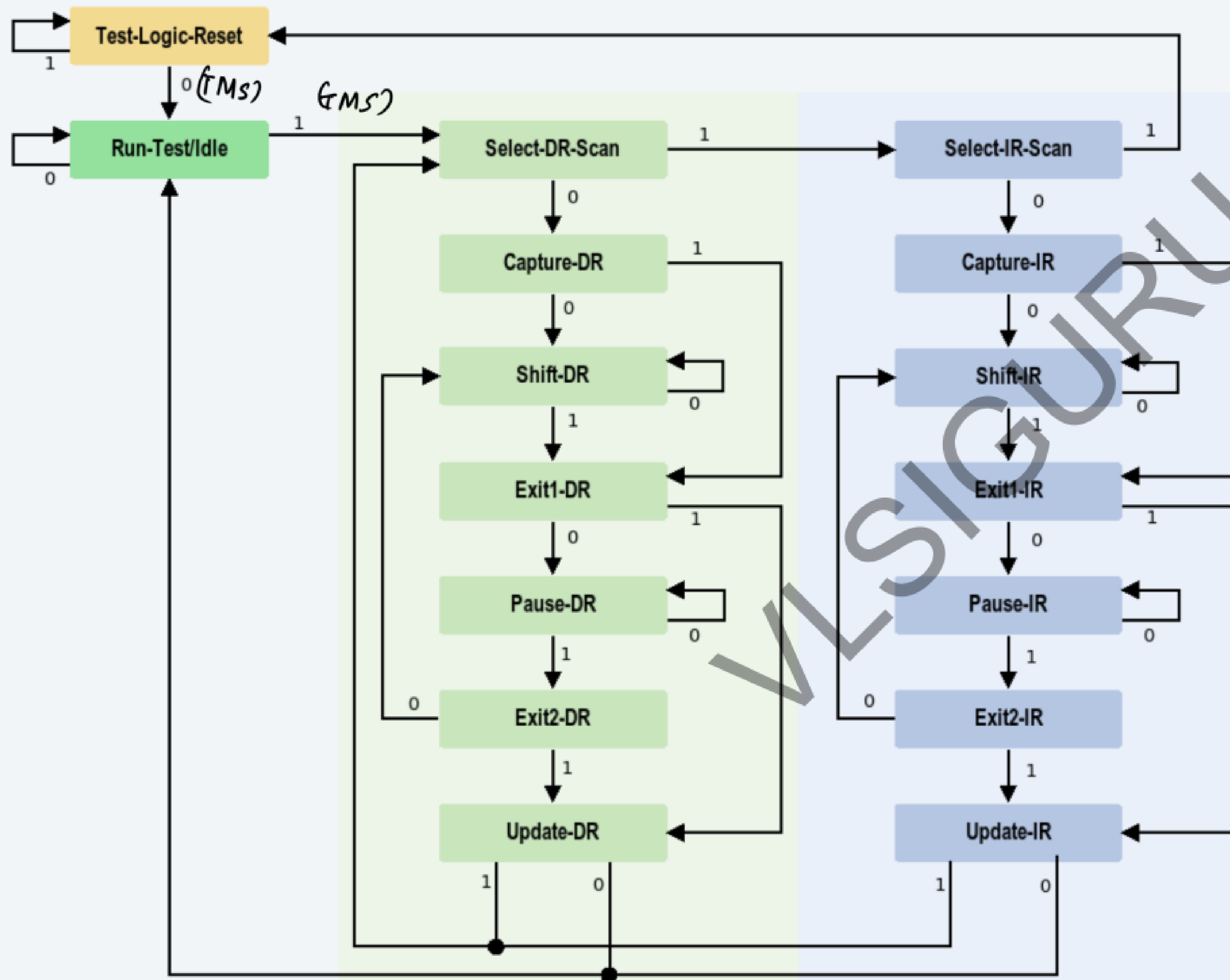
Selects one particular data register based on the instruction loaded into the instruction register.

Select line : Output of instruction register will be given to a decoder. The output of decoder acts as the select line of mux B.

Need of Decoder :

To reduce the no. of bits of the instruction register's shift register and hold register.

TAP CONTROLLER



16 State FSM

Consists of

- Reset State
- Run-Test/Idle
- Two major branches
 - Access to IR
 - Access to DR

Test-Logic-Reset state :

- Test logic is disabled
- Normal operation of the chip can continue

FSM enters this state

- i. When TRST_N is pulled active ($\text{TRST_N} = 0$)
(or)
- ii. When TMS input is held active for at least 5 clock cycles
- iii. Upon power up (process of applying power to the entire chip [including the JTAG module]. When power is applied, JTAG FSM enters the rest state)

Run-Test/Idle:

This is a state in which the FSM is waiting for some test operations to complete.

Example :

If we are going to do core-logic test for a block, then the following signals (int_mode, ext_mode, int_ltest_en, ext_ltest_en)will be controlled to their appropriate value through TDR to start the operations.

Till the time core-logic test completes, the FSM will be in the Run-Test/Idle state.

Select-DR/Scan and Select-IR/Scan

Temporary state to initiate the test data sequence for IR/DR respectively.

Capture-IR :

The shift register contained in the instruction register loads a fixed value (two LSB are 0 and 1)

This is done to check whether there is any SA0 and SA1 fault in the TDI and TDO line.

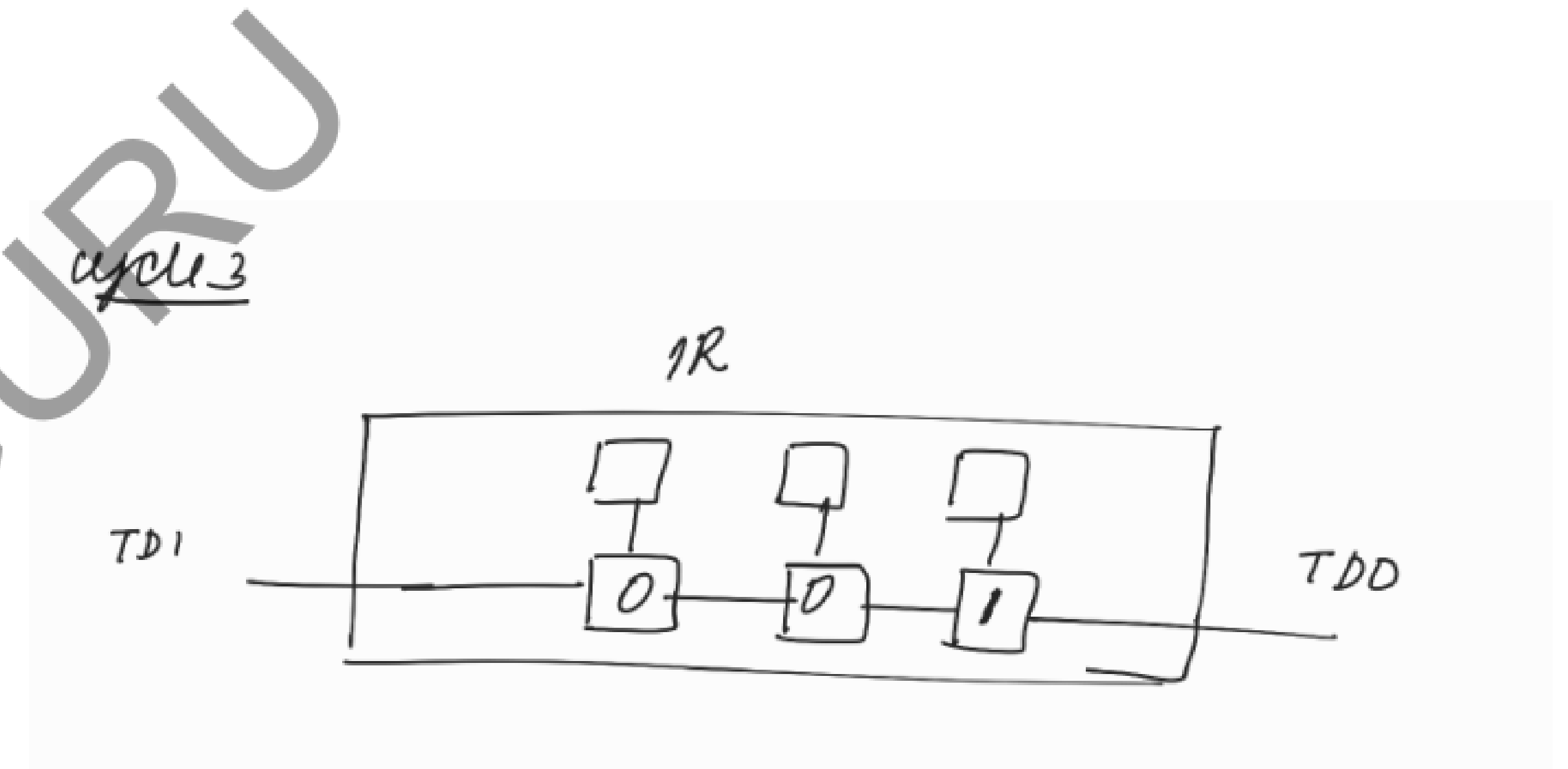
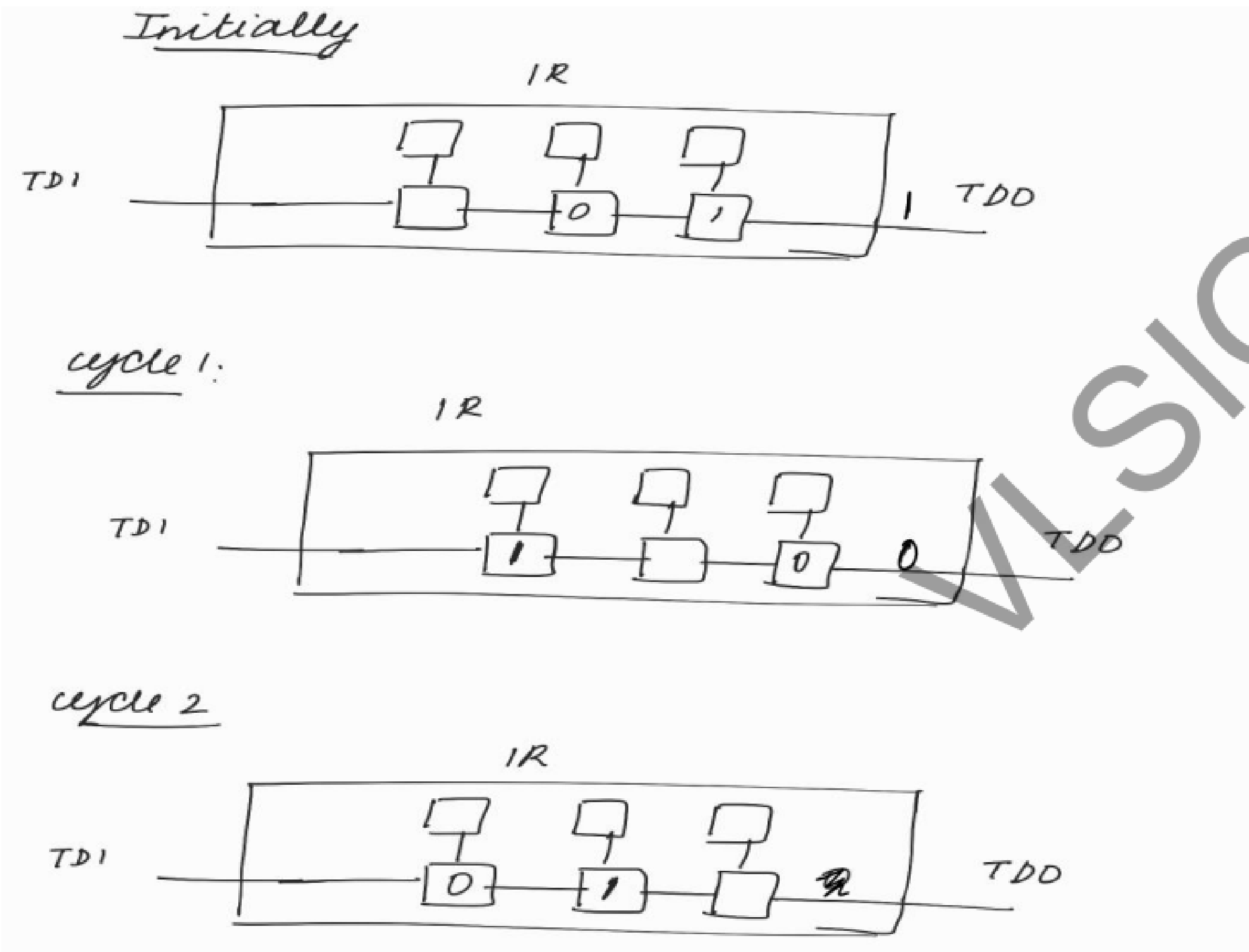
Shift-IR :

During this state, the required instruction will be serially loaded into the shift register inside the instruction register.

For each clock cycle (TCK), one bit is shifted in and out of instruction register through TDI and TDO.

ILLUSTRATION :

Suppose if Data reg 1 is to be selected and the instruction to select DR1 is 001, then the required instruction will be loaded into the IR's shift register during shift IR



Note :

Explanation of no. of bits in the IR :

3 mandatory instructions : Exttest, sample preload, Bypass

IR must be at least 2 bit wide in order to select these mandatory instructions.

The size of IR can be increased based on the no. of TDRs we are having.

Exit1- IR :

A temporary state

TMS is held high, the controller enters the update-IR state which terminates the scanning process.

Pause-IR :

Allows shifting of IR to be temporarily halted.

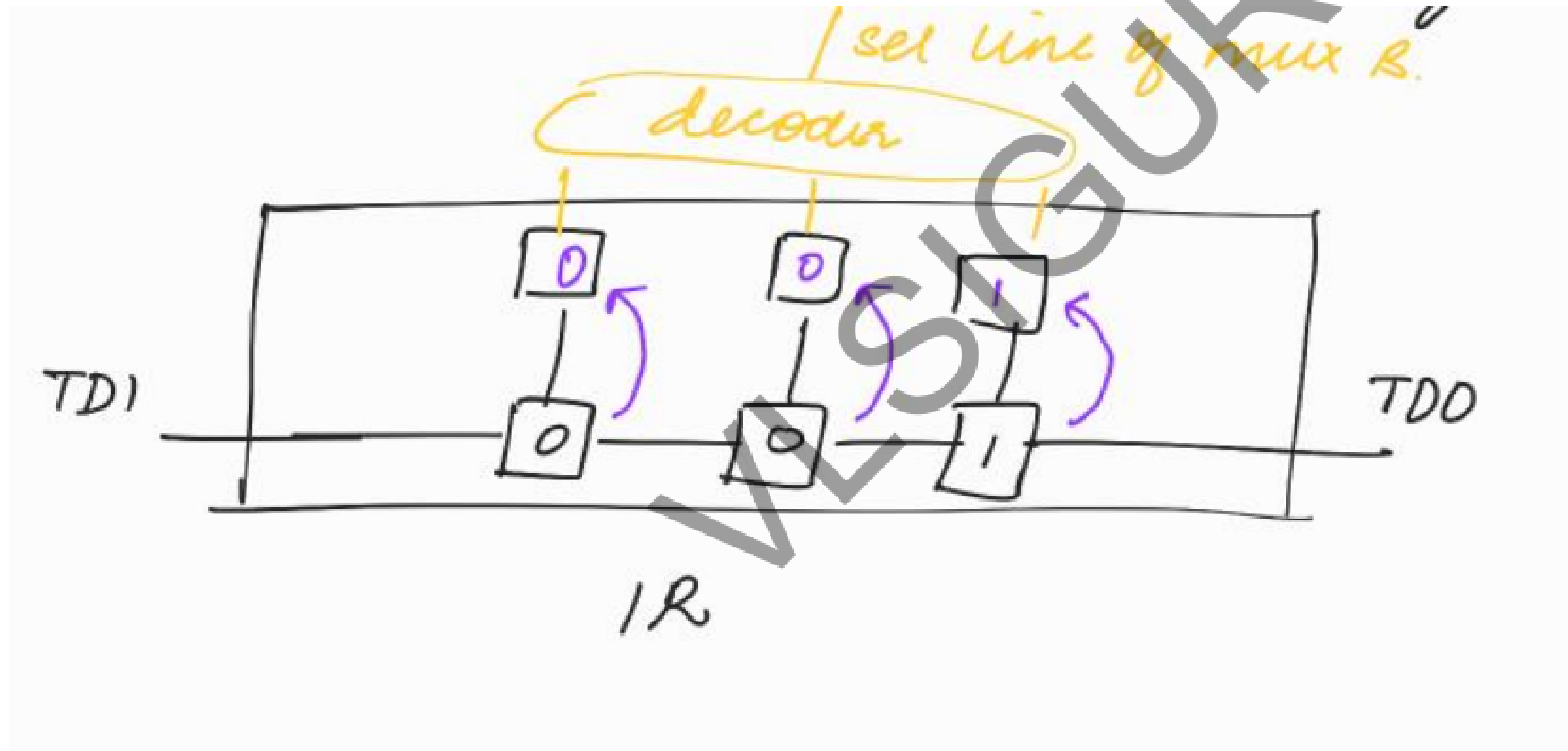
Exit2- IR :

A temporary state

TMS is held high, the controller enters the update-IR state which terminates the scanning process.

Update-IR :

The instruction which has been shifted into the IR's shift register is parallelly loaded into the IR's hold register.



Capture-DR :

Data may be parallelly loaded into the test data register that is selected by the current instruction.

Shift-DR :

During this state, the required data will be serially loaded into the shift register inside the Data register.

For each clock cycle (TCK), one bit is shifted in and out of data register through TDI and TDO.

If DR1 is used for core test control

It controls these signals : int_mode, ext_mode, int_ltest_en, ext_ltest_en

Note : No.of bits of DR will be equal to the no. of signals that particular DR is controlling.

Suppose we are testing the core-logic test of a block:

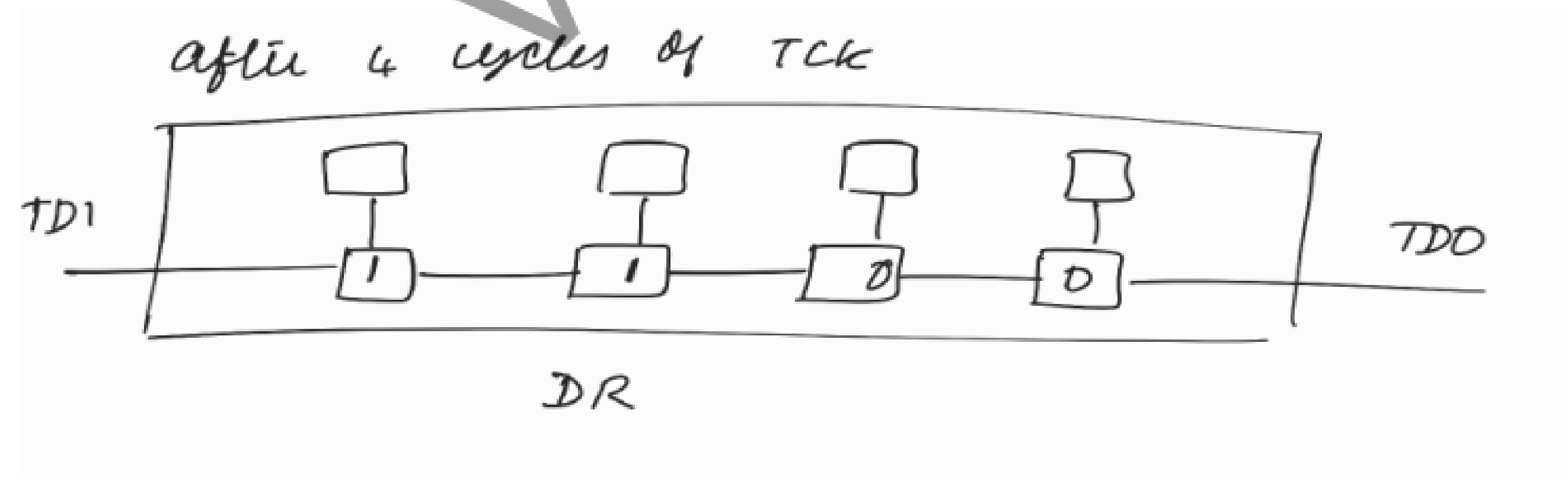
Then int_mode = 1

ext_mode = 0

int_ltest_en = 1

ext_ltest_en = 0

After 4 cycles of TCK



Exit1- DR :

A temporary state

TMS is held high, the controller enters the update-DR state which terminates the scanning process.

Pause-DR :

Allows shifting of DR to be temporarily halted.

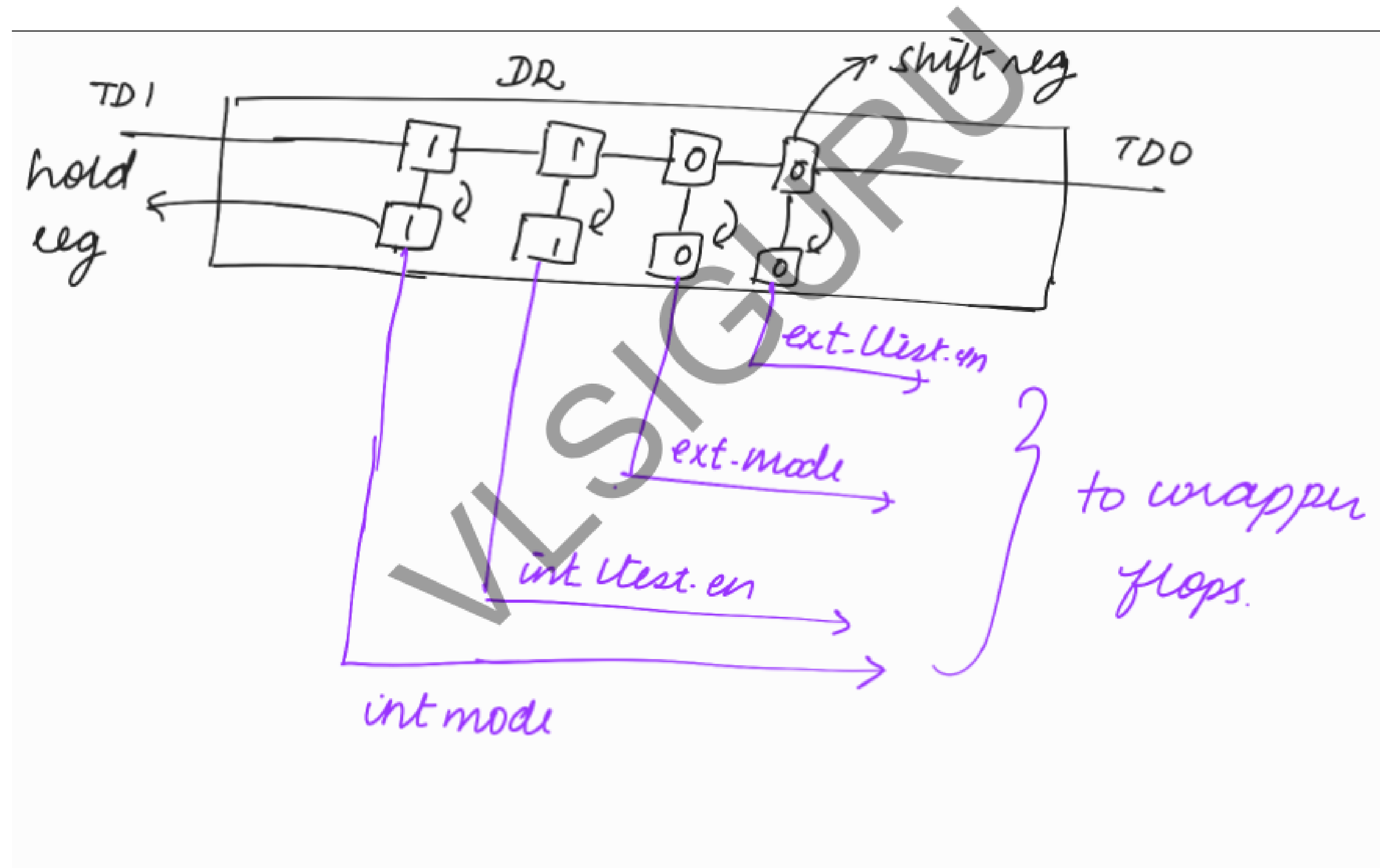
Exit2- DR :

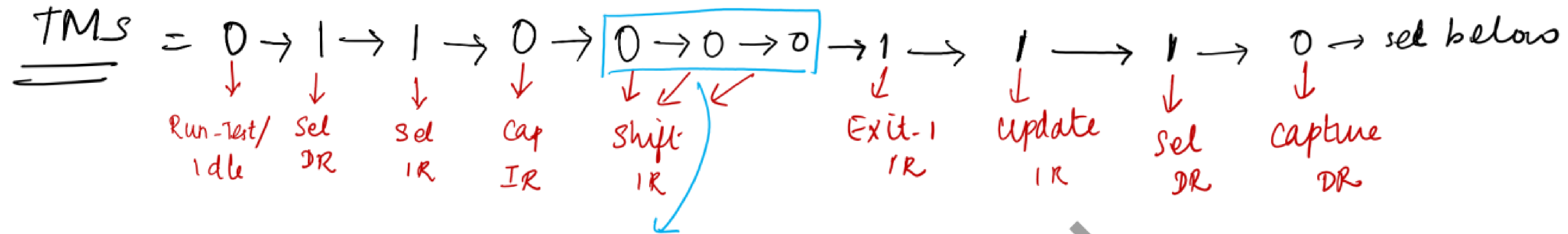
A temporary state

TMS is held high, the controller enters the update-DR state which terminates the scanning process.

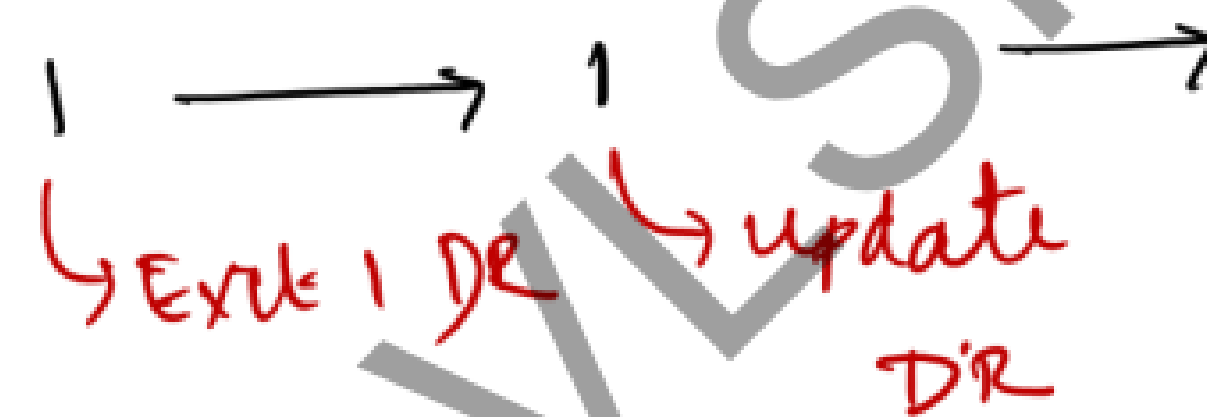
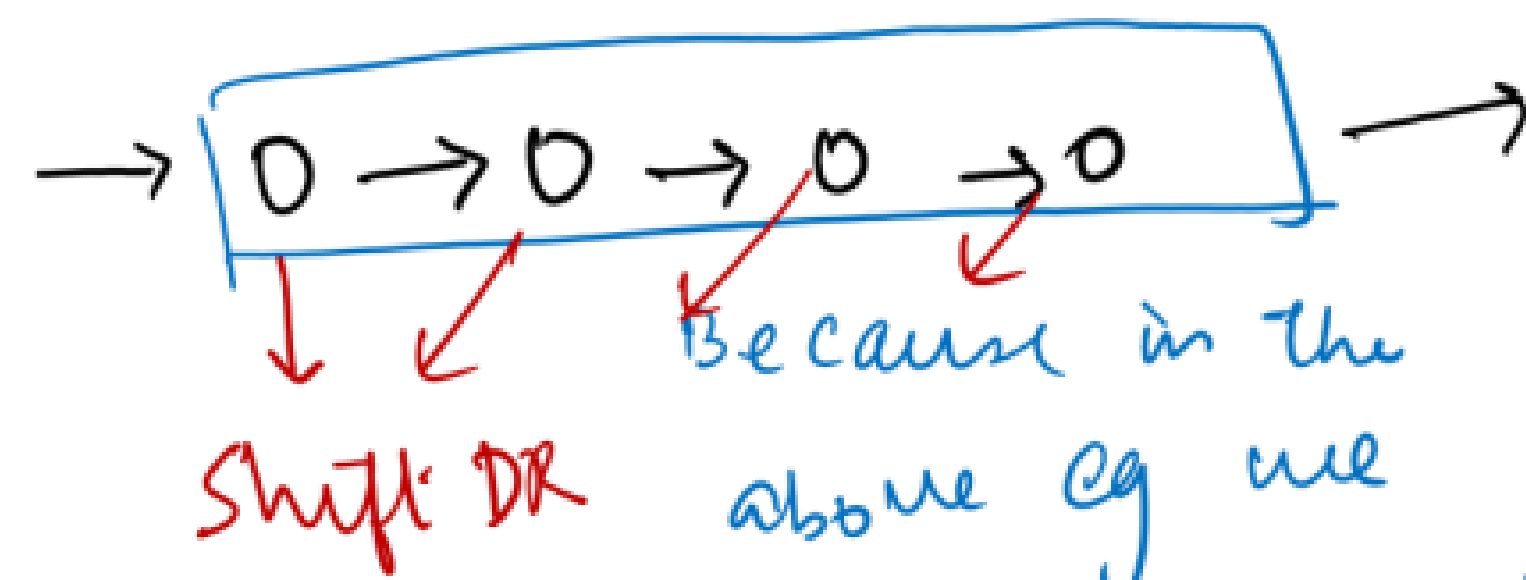
Update-DR :

The data which has been shifted into the DR's shift register is parallelly loaded into the DR's hold register.





Because in the above eg we are loading 3 bit ins. Till the time ins is loaded into the IR's shift reg the FSM has to be in shift IR state



After this TMS will be maintained 0 till the time the operation is completed so that the FSM can remain in the runtest/Idle

0 $\rightarrow$ RunTest/Idle

This will initiate the operation. Till the particular operation completes, FSM will be there in the runtest/Idle.